National University of Computer and Emerging Sciences, Lahore Campus				
	Course:	**Machine Learning for Robotics**	**Course Code:**	**CS 5096**
	Program:	**BS (Computer Science)**	**Semester:**	**Spring 2026**
	Date:	**09-02-2026**	**Total Marks:**	**20**
	Section:	**A**	**Weight:**	
	Exam:	**Quiz 1**	**Roll No:**	

Q1) Use a single iteration of gradient descent to update the weights of a linear regression model on the following dataset. Let $\theta_0 = 0.5$, $\theta_1 = 0.5, \alpha = 0.1$ initially. The cost function is given by: **(13)**

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})^2$$

Find the updated values of θ_0 and θ_1 after this iteration.

Driving Experience (years)	Monthly Auto Insurance Premium ($)	$h_\theta(x^{(i)})$	$h_\theta(x^{(i)}) - y^{(i)}$	$(h_\theta(x^{(i)}) - y^{(i)})x$
5	64	3	-61	-305
2	87	1.5	-85.5	-171
12	50	6.5	-43.5	-522
		$\sum_{i=1}^{m}$	-190	-998

$$\frac{\partial J}{\partial \theta_0} = \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})$$

$$\frac{\partial J}{\partial \theta_1} = \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})x^{(i)}$$

$\theta_i = \theta_i - \alpha \frac{\partial J}{\partial \theta_i}$

$\theta_0 = 0.5 - 0.1(-190/3) = 6.833$

$\theta_1 = 0.5 - 0.1(-998/3) = 33.767$

1 mark for each correct value

Q2) Use the normal equation to find the exact weights for the dataset in Q1. **(7)**

Let $X = \begin{bmatrix} 1 & 5 \\ 1 & 2 \\ 1 & 12 \end{bmatrix}$, $y = \begin{bmatrix} 64 \\ 87 \\ 50 \end{bmatrix}$, and $X^T X = \begin{bmatrix} 3 & 19 \\ 19 & 173 \end{bmatrix}$.

1 mark for determinant
1 mark for inverse
2 marks for each correct successive matrix multiplication
1 mark for final weights

$$X^T X = \begin{bmatrix} 1 & 1 & 1 \\ 5 & 2 & 12 \end{bmatrix} \begin{bmatrix} 1 & 5 \\ 1 & 2 \\ 1 & 12 \end{bmatrix} = \begin{bmatrix} 3 & 19 \\ 19 & 173 \end{bmatrix}$$

$$\det(X^T X) = (3)(173) - (19)(19) = 519 - 361 = 158$$

$$(X^T X)^{-1} = \frac{1}{158} \begin{bmatrix} 173 & -19 \\ -19 & 3 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 1 & 1 & 1 \\ 5 & 2 & 12 \end{bmatrix} \begin{bmatrix} 64 \\ 87 \\ 50 \end{bmatrix} = \begin{bmatrix} 201 \\ 1094 \end{bmatrix}$$

$$\theta = (X^T X)^{-1} X^T y = \frac{1}{158} \begin{bmatrix} 173 & -19 \\ -19 & 3 \end{bmatrix} \begin{bmatrix} 201 \\ 1094 \end{bmatrix}$$

$$= \frac{1}{158} \begin{bmatrix} 173(201) - 19(1094) \\ -19(201) + 3(1094) \end{bmatrix} = \frac{1}{158} \begin{bmatrix} 13987 \\ -537 \end{bmatrix}$$

$$\theta = \begin{bmatrix} 88.53 \\ -3.40 \end{bmatrix}$$